



MOM!

You're Out of This World!

India made history with its Mars Orbiter Mission this year. We explore the innovation, technology and significance of this great accomplishment.

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Zero hour, four seventeen a.m. 24 September 2014. It had been 298 days since the launch of India's Mars Orbiter Mission or Mangalyaan, marking the first time India would venture beyond the Moon into the orbit of another celestial body. Scientists, engineers and researchers at the ISRO's Missions Operations Complex (MOX) in Bangalore were gathered together awaiting the big news. Joined by their colleagues across the world in support stations at Lucknow, Mauritius, Indonesia and Norway they knew the moment of truth was at hand. Will the indigenously designed Liquid Apogee Motor engines on the PSLV rocket awaken after nearly 10 months in dormancy? Would this moment lead India into the record books? Or will the failure be just another footnote in history?

Seconds later as the confirmation of the engines activating reached the MOX, the burst of applause and cheering was deafening. History had been made. India became the first nation to reach Mars orbit on its first attempt and the first Asian nation to reach interplanetary space, beating out China and Japan. In addition to this the Indian Space Research Organisation or ISRO joined the world leaderboards as the fourth space agency, after NASA, the Soviet space program and the European Space Agency, to reach Mars orbit. This great accomplishment came at the end of many years of hard struggle and innovation for everyone involved, and perhaps the first step in to a new era of Indian space flight.

A Journey of 414 Million Miles Begins

The roots of India's relationship with space exploration hasn't only been one of pure discovery, but also of survival. Before we

could think to go to the Moon, much less Mars, it was more important to feed the hundreds of millions in the country. In the post-Independence phase of Indian growth, the need to leverage every technological and scientific means to attain self-sustenance was critical. The 1980 launch of the Rohini satellite on the Satellite Launch Vehicle 3 (SLV-3) was a momentous occasion since India had successfully created its own launch vehicle within 11 years of the creation of the Indian Space Research Organisation. What had begun in 1969 with the help of the Soviets, evolved to become an indigenous aerospace industry catering to the needs of the world.

Just over thirty years ago, in April 1984, India made its first historic foray into space travel. With the help of the Soviet Intercosmos Program, Squadron Leader Rakesh Sharma flew aboard the Soyuz T-11 rocket to the Soviet Salyut 7 Orbital Station. Gaining instant national recognition, Rakesh Sharma became the first Indian cosmonaut in space. Finally, an Indian could see the world as it really was and report it first hand. So when the Prime Minister at the time, Indira Gandhi asked him how India looked from space, he fittingly replied - "Sare jahan se achha...".

There are many such stories that demonstrate the indomitable spirit of Indian innovation. But very few face such a long journey to fruition. Beyond parochial definitions of patriotism and nationalist spirit, India has always been about scientific discovery. The curiosity that led Aryabhata to discover natural cosmic phenomena and Satyendra Nath Bose make discoveries in experimental quantum mechanics was the same curiosity that led India to take on the seemingly impossible challenge of reaching



MOM's proud parents